



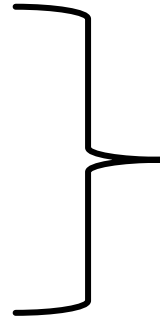
ELECTRICAL MACHINES

Instruction

Lecture (CM): 14h

Exercise (TD): 14h

Practical (TP): 8h



Total: 36h

Evaluation

Attendance: XX%

Mid-term exam: XX%

Final exam: XX%

Solving Exercise (TD): XX%

Report for TP: XX%

Outline

- Section 1: AC-circuit + Machinery Principle**
- Section 2: Induction Motor (include TP)**
- Section 3: Synchronous Motor (include TP)**
- Section 4: Synchronous Generator**
- Section 5: DC Machines (include TP)**



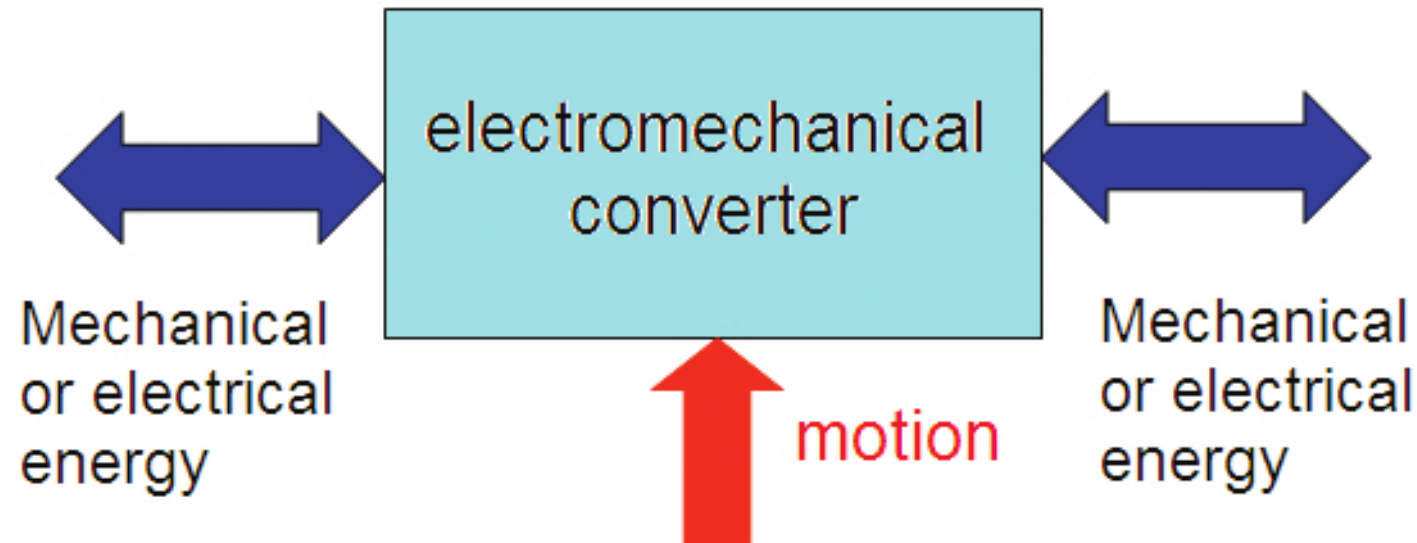
Section 1

AC-CIRCUIT + MACHINERY PRINCIPLE

I. Introduction to electromechanical energy conversion

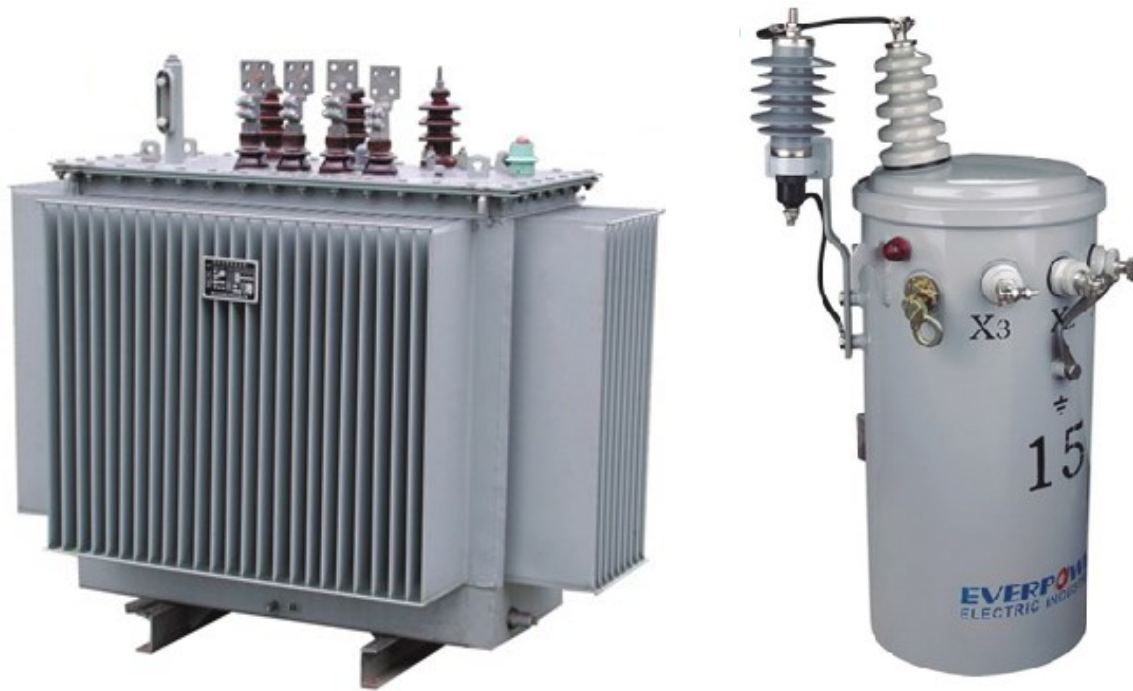
a. What is electromechanical energy conversion?

Electromechanical energy conversion is a **conversion** of mechanical energy into electrical energy (generator) or vice versa (motor) with the aid of rotary motion (rotary machines).



Electromechanical energy conversion

Transformers and **converters** do not belong to the group of electromechanical energy conversion devices because they only convert one kind of electrical energy into another kind of electrical energy with different parameters (change in voltage, current, frequency, number of phases, conversion of DC into AC current, etc.) without any motion.



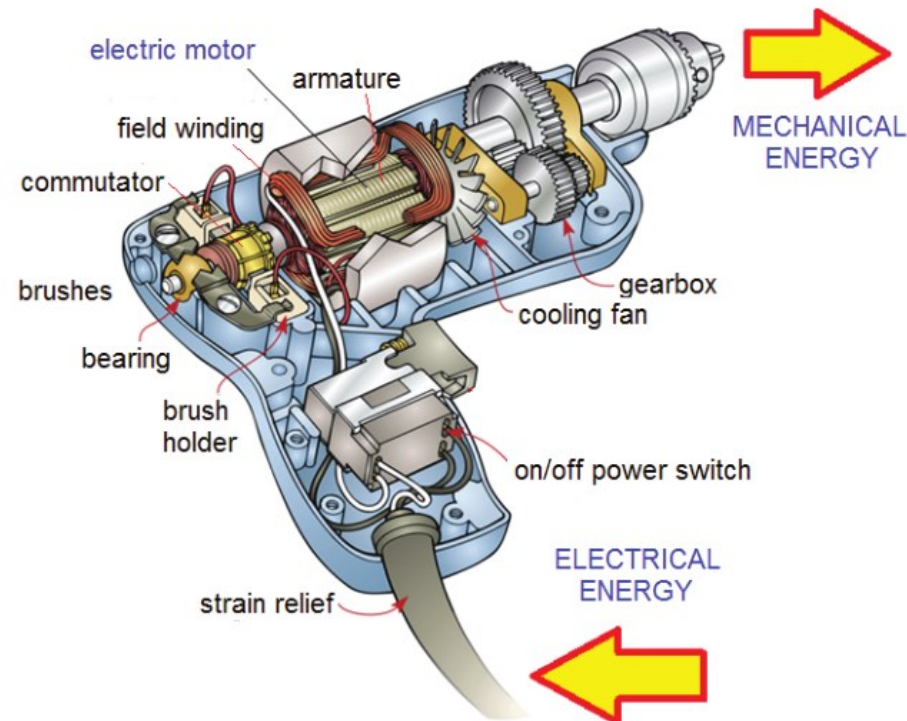
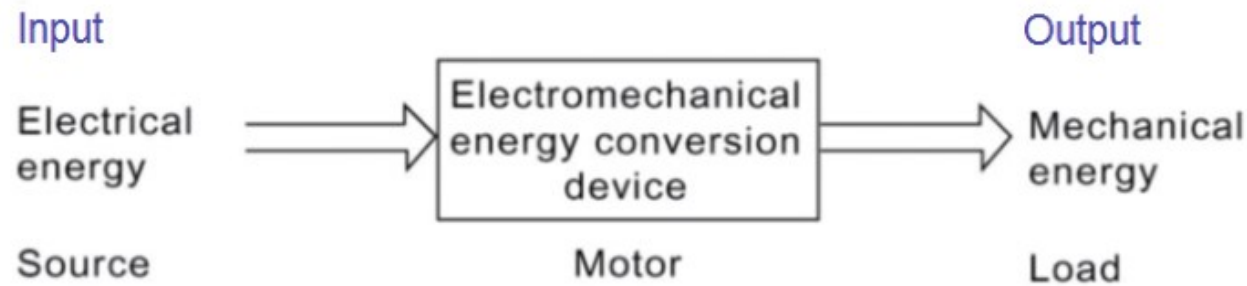
Transformers



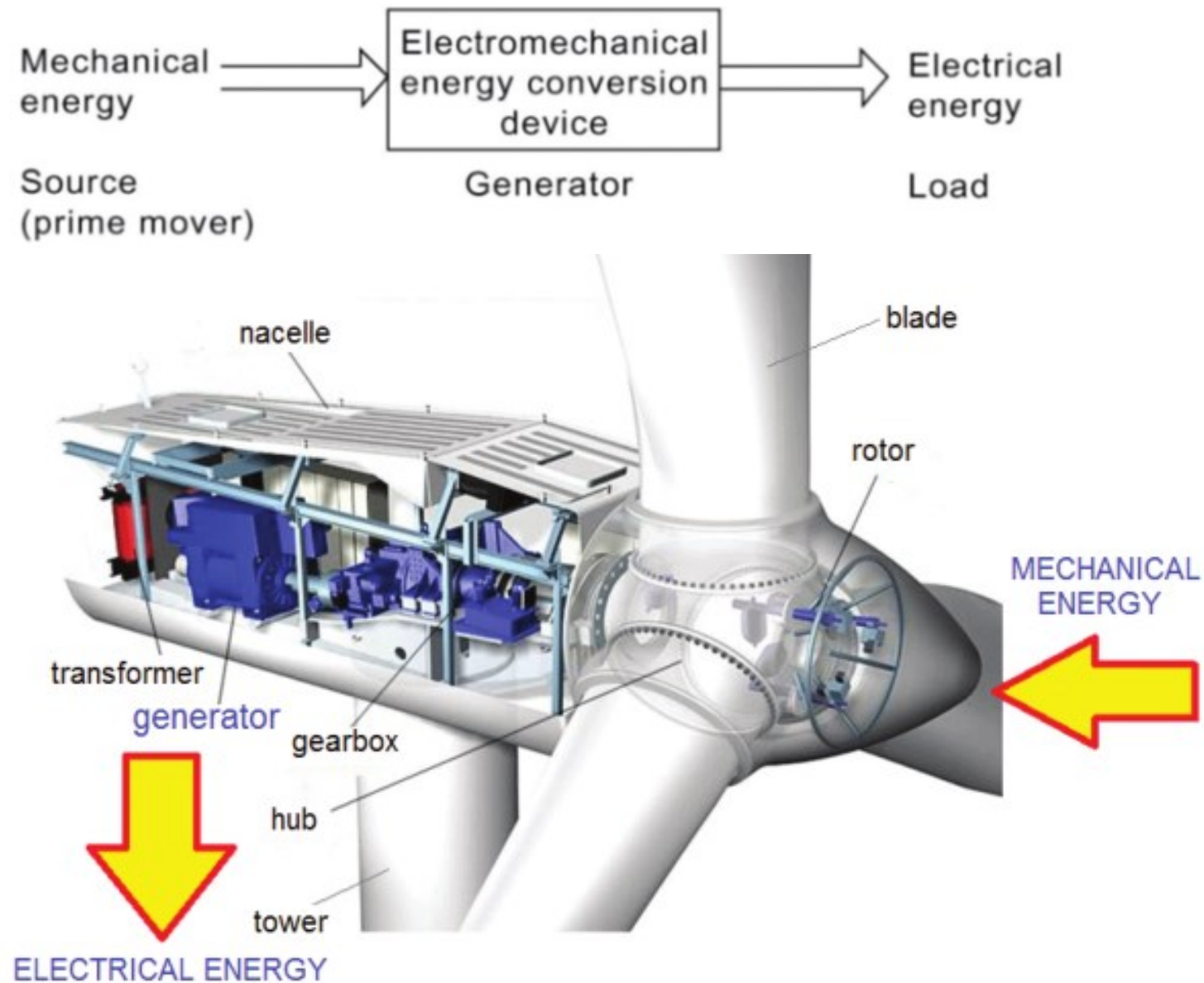
Converters

b. Block diagrams of electromechanical energy conversion devices

An example of application of an electric motor is:



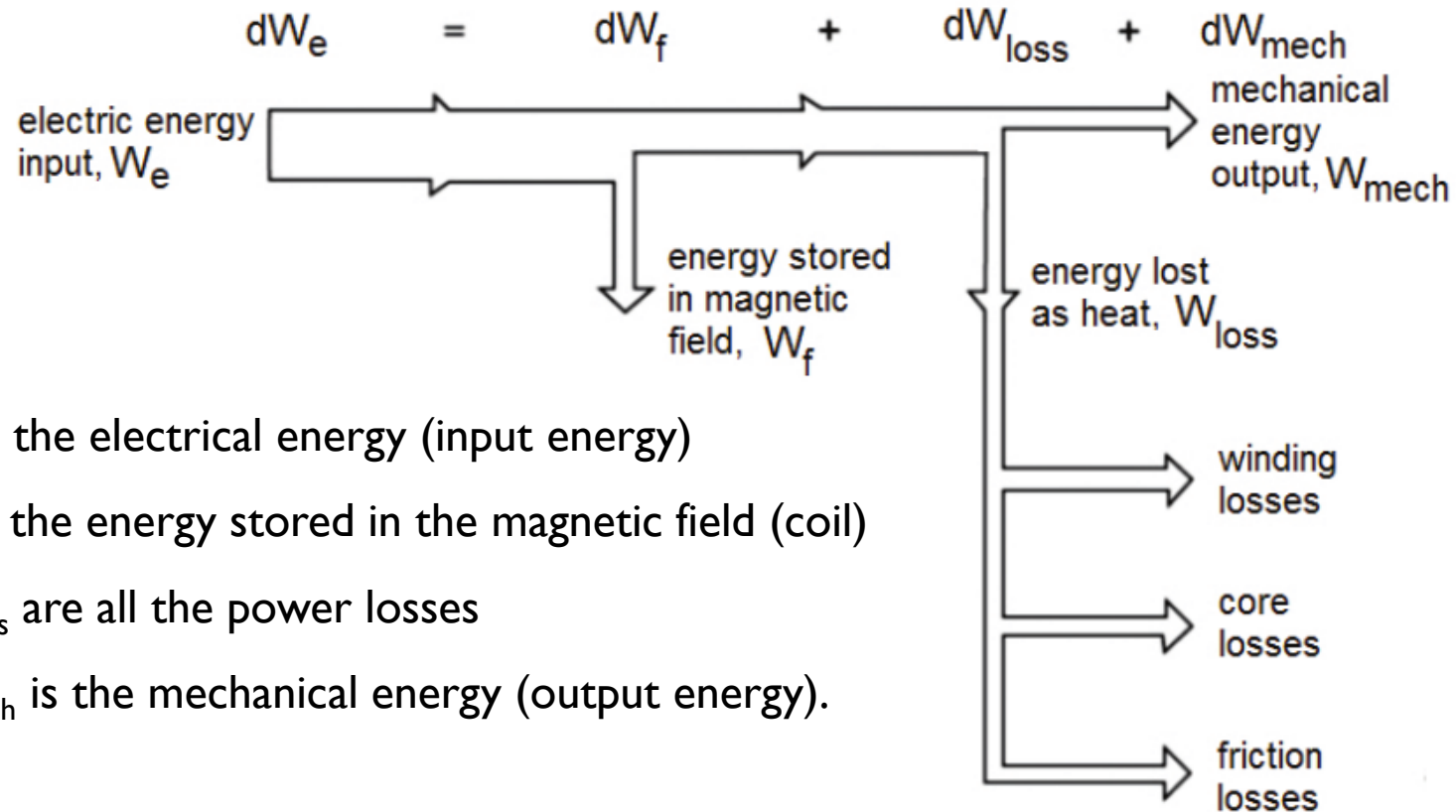
An example of application of an electric generator:



c. Energy flow in an electromechanical energy conversion device

Figure below illustrates the conversion of electrical energy into mechanical energy, according to the following equation:

$$dW_e = dW_f + dW_{loss} + dW_{mech}$$

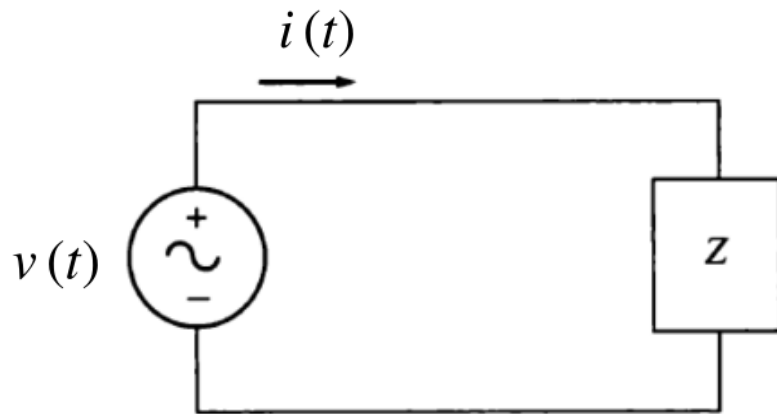


Where: dW_e is the electrical energy (input energy)
 dW_f is the energy stored in the magnetic field (coil)
 dW_{loss} are all the power losses
 dW_{mech} is the mechanical energy (output energy).

d. Review of ac-circuit

- **Single-phase circuit:**

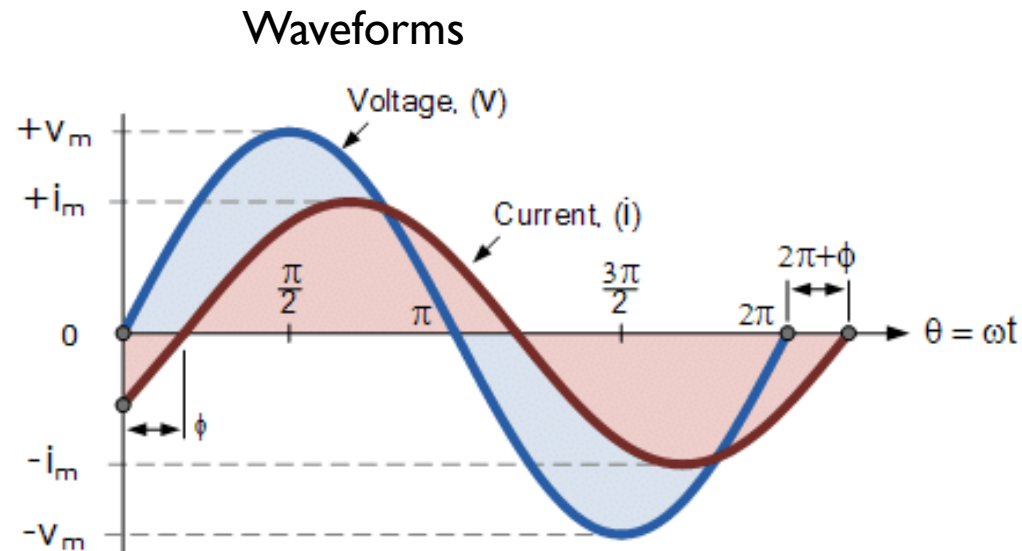
In electrical power systems, we are mainly concerned with flow in the electrical circuit, such as Voltage (V), Frequency (f), Current (I), and Power (P).



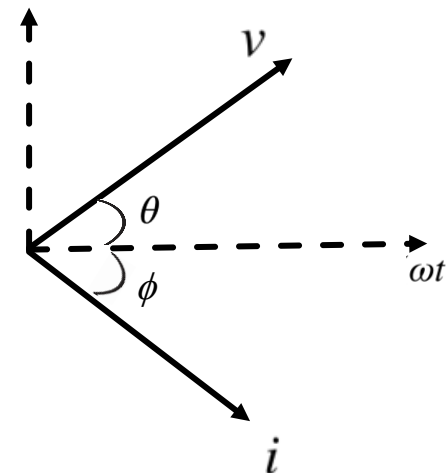
General form:

$$v(t) = V_m \cos(\omega t + \theta)$$

$$i(t) = I_m \cos(\omega t + \theta - \phi)$$



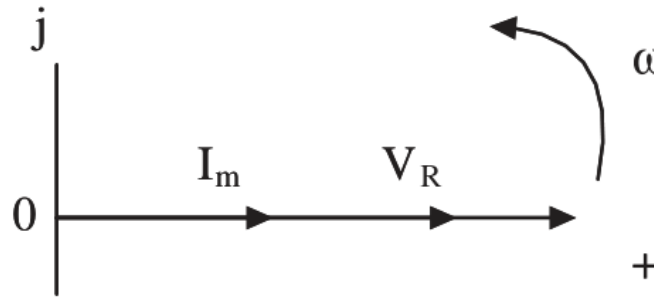
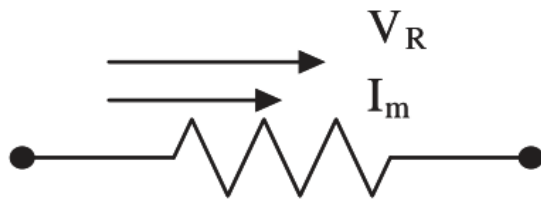
Phasor diagram



- **Single-phase circuit:**

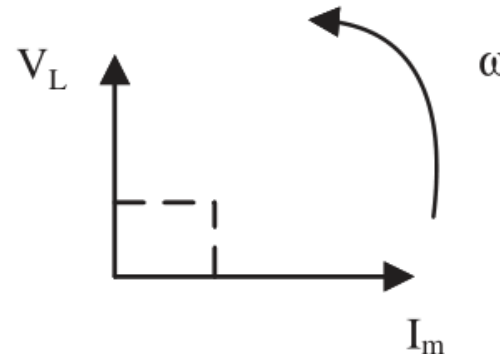
For impedance, if the circuit is purely resistive, inductive, or capacitive, there will be difference in current and angle phase shift.

- **Purely Resistive Circuit:**



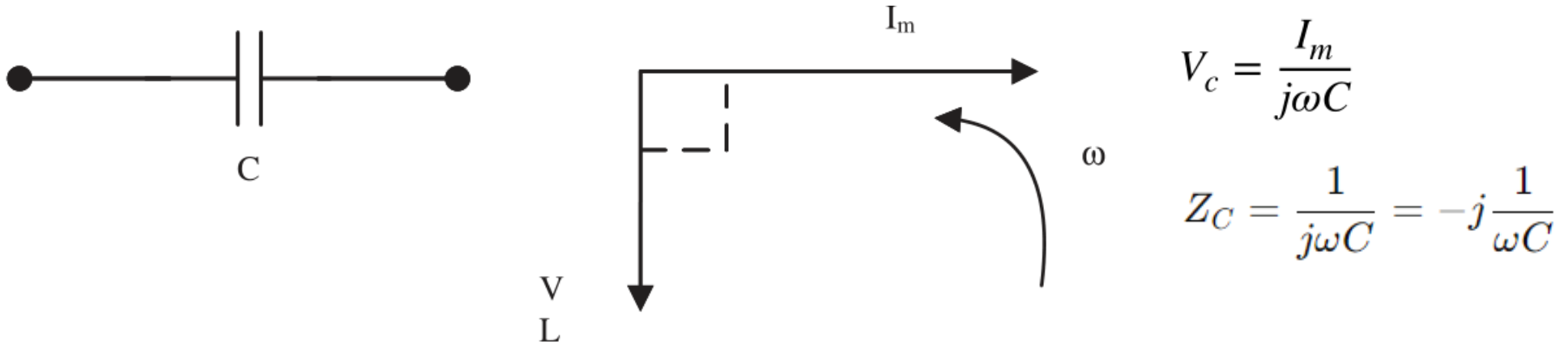
$$V_R = RI_m.$$

- **Purely Inductive Circuit**

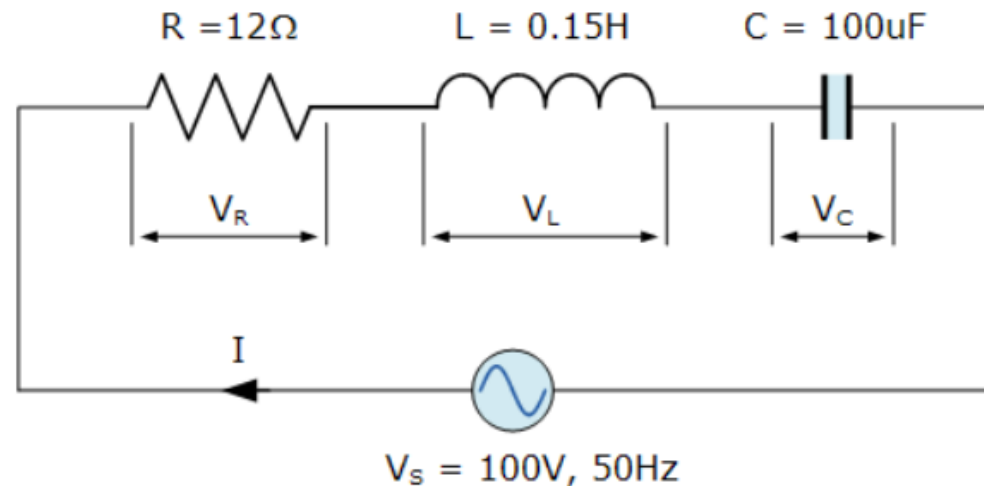


$$V_L = j\omega LI_m. \quad Z_L = j\omega L$$

- **Single-phase circuit:**
 - **Purely Capacitive Circuit:**

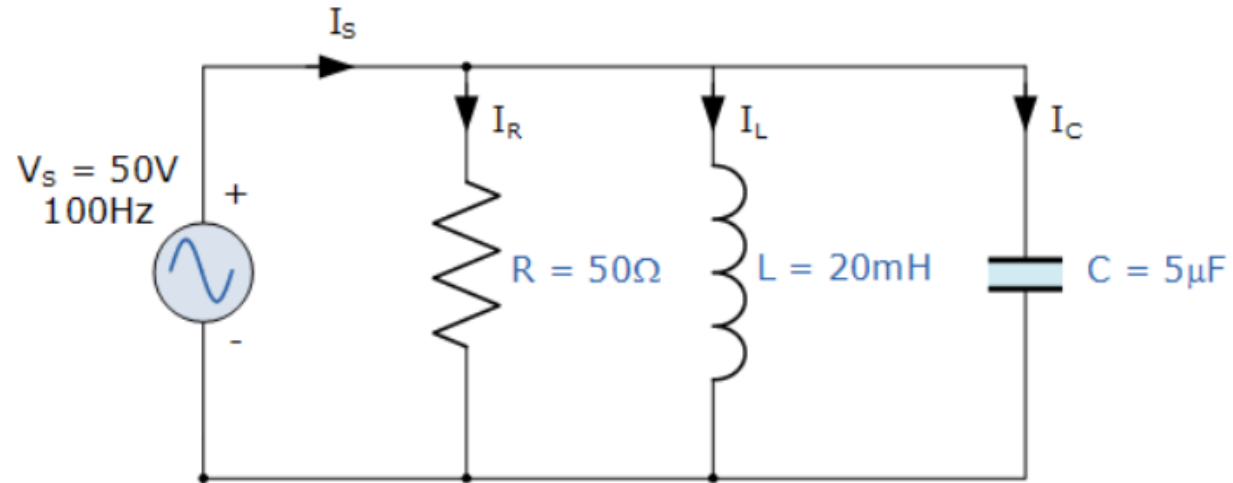


Example 1: Calculate the impedance of a series RLC circuit

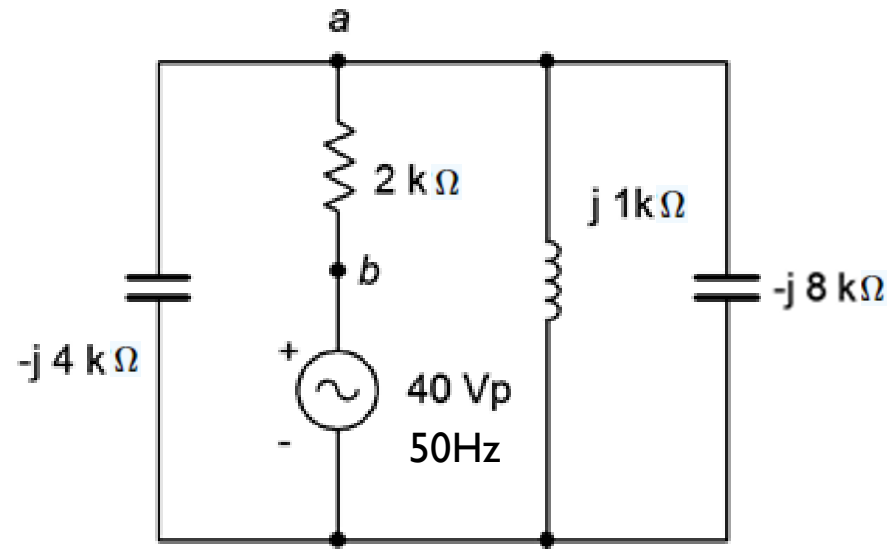


$$Z_L = j\omega L$$

Example 2: Calculate the impedance of a parallel RLC circuit:



Example 3: Calculate the impedance of a RLC circuit:



• Powers and power factor:

- **Active power:** $P = V_{rms} I_{rms} \cos \varphi$

$$V_{rms} = \frac{V_m}{\sqrt{2}}$$

$$I_{rms} = \frac{I_m}{\sqrt{2}}$$

- **Reactive power:** $Q = V_{rms} I_{rms} \sin \varphi$

- **Apparent power:** $S = V_{rms} I_{rms}$

- **Power factor:** $\cos \varphi = \frac{\text{Active Power}}{\text{Apparent Power}} = \frac{P \text{ (KW)}}{S \text{ (KVA)}}$